

11	(a) X-ray: flat/shadow/2D image regardless of depth of object/depth not indicated	B1	B1	
	CT scan: built up from (many) images at different angles image is three-dimensional image can be rotated/viewed at different angles	B1	B1	[5]
(b)	(i) $I = I_0 e^{-\mu x}$ $0.25 = e^{-0.69x}$ $x = 2.0 \text{ mm}$ (<i>allow 1 s.f.</i>)	C1	A1	[2]
	(ii) for aluminium, $I/I_0 = e^{-0.46 \times 2.4}$ $= 0.33$ fraction $= 0.33 \times 0.25$ $= 0.083$	C1	A1	[2]
	(iii) gain/dB $= 10 \lg(I/I_0)$ $= 10 \lg(0.083)$ $= (-) 10.8 \text{ dB}$ (<i>allow 2 s.f.</i>) with negative sign	C1	A1	B1
				[3]